

LEIN PASACHE

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EDUCATION

California State Polytechnic University, Pomona (CPP) – M.S. MECHANICAL ENGINEERING | GPA: 3.33 **Expected:** June 2026
University of California, Los Angeles (UCLA) – B.S. MECHANICAL ENGINEERING | GPA: 3.66 June 2024
Dean's Honor List 2023, Engineering Achievement Award in Student Welfare 2024

SKILLS AND RELEVANT COURSES

Technical Mechanical Design, Geometric Dimensioning & Tolerancing, Prototyping, Data Analysis, Cleanroom Assembly
Hardware Additive Manufacturing/3D Printers, Soldering, Basic Machine Shop Tools
Software SolidWorks, Siemens NX, Teamcenter, AutoCAD, MS PowerPoint/Excel/Office/SharePoint, AirTable
Coding MATLAB, Python
Courses Electrical Circuits I, Intro to Aeronautics, Intro to Space Technology, Spacecraft Design
Languages English and Spanish

INTERNSHIP EXPERIENCES

NASA Jet Propulsion Laboratory (JPL) – ENGINEERING UNDERGRADUATE/GRADUATE STUDENT INTERN *Sept. 2022 – Present*
Engineer Intern, NEO Surveyor (Near Earth Object Surveyor), Enclosure Team

- Designed MGSE for thermal strap installation on spacecraft radiators, ensuring cleanroom compatibility and efficient assembly.
- Verified safety with 2.5 factor of safety analysis, avoiding physical testing and meeting standards.
- Delivered complete drawing packages (parts, subassemblies, assemblies) to guide successful fabrication and integration.
- Procured COTS components that met performance specs, reducing lead time and simplifying assembly logistics.

Intern, MSR (Mars Sample Return), Orbiting Sample

- Designed low-cost, rapid manufactured hardware for Mechanical Ground Support Equipment (MGSE) and tooling for assembly.
- Developed procedures for testing, usage, and set-up of MGSE design for non-Flight Integration and Testing.
- Produced CAD models and engineering drawings using NX. Additionally analyze data from thermal test using MATLAB.
- Reduced the assembly process with MGSE and tooling by 90% (from 4 hrs. to 20 min), submitted progress to the team.

Integration and Testing Engineer, FORGE (Fluidic Operations in Reduced Gravity Experiment)

- Managed integration build flow and test operations of FORGE troubleshoot Python script to identify the optimal time frames.
- Updated NX model of light panel and observation structure after mechanical modifications.
- Designed pull-out and pull-through tests for metal inserts in 3D-printed part.
- Executed test successfully and determined that the expected loads on the inserts were below its failing limits by 70%.

Performance Team a Maersk Company – ENGINEERING INTERN

Jun. 2022 – Sept. 2022

- Edited existing CAD drawings of warehouse racking with guidance from Mechanical Engineers.
- Visited sites to supervise racking assembly process and take measurements to revise existing rack layouts.
- Charted CAD warehouse layouts for project proposals, racking, fire and egress locations, and architectural drawings.
- Created dynamic blocks for racking design and elevations that shortened the design time by 50%.

PROJECTS

Capstone Project: Autonomous Delivery Rover

Jan. 2024 – Jun. 2024

- Designed and built an autonomous rover that navigated a closed loop track picking up 3 objects and delivering them: placed 1st.
- Spent one quarter doing testing and design reviews and a second quarter building and troubleshooting the rover.

Specialized Spoon for Assisting People with Diminished Hand/Finger Dexterity to Retrieve Medication

Jan. 2024 – Mar. 2024

- Optimized a design to leverage additive manufacturing strength by incorporating pre-assembled components and complex geometry
- Wrote a report on lessons learned after prototyping 3 iterations using FDM printers, also added future developments for the product.

Spacecraft Design for Mock Europa Sample Return Mission

Apr. 2023 – Jun. 2023

- Collaborating to define level 1-3 requirements, do risk assessment, design subsystems, and perform trade studies.
- Leading structure subsystem design through use of CAD software to model spacecraft components and structure.

EXTRACURRICULAR

Northrop Grumman Collaboration Project (NGCP) – MANUFACTURING DEPUTY

Aug. 2024 – Present

American Institute of Aeronautics and Astronautics (AIAA) – STUDENT MEMBER

Oct. 2021 – Present

Society of Latinx Engineers and Scientists (SOLES) – STUDENT MEMBER

Oct. 2020 – Present

Society of Women Engineers (SWE) – STUDENT MEMBER

Oct. 2020 – Present

Design/Build/Fly (DBF) at UCLA – STUDENT MEMBER

Feb. 2022 – Jun. 2024